

# Database Management System

NAME \_\_\_\_\_  
DATE: / /

## Unit I: Introduction

Database and Database Users, characteristics of the Database Approach, Different People Behind DBMS, Implication of Database Approach, Advantages of Using DBMS, When not to use a DBMS

② Database System concepts and Architecture: Data models, Schemas and Instances, DBMS architecture and Data Independence, Data Base Languages and Interface, The Database system environment, classifications of database Management Systems

③ Data Modeling using the Entity Relation Model: High level conceptual data models for database design with an Example, Entity Types, Entity sets, Attributes and Keys, ER-Model concept, Notations for ER Diagrams, Proper Naming of constructs, Relationships Types of Degree than two.

## Unit II:

Relational Data Model and Relational Algebra

Relational Model concepts, Relational Model constraints and Relational Database schema, Defining relations, Update operations on Relations and constraint

violations Basic Relational Algebra operations

Additional Relational operations, Queries in relational algebra using all the operations

## Unit III

Functional Dependencies and Normalization for Relational Databases

Informal Design Guidelines for Relational schemas, Functional Dependence, Normal Forms Based on primary keys, General Definitions of second and Third Normal Forms, Boyce - Codd Normal Form (BCNF)

Unit 4  
Relational Database Language: Data Definition

In SQL - Queries in SQL, INSERT, DELETE, UPDATE statements SQL,

Data Type in SQL: Number Types, character Types

## Unit IV

Components of SQL

Data Definition Language (DDL), Data Manipulation Language (DML), Query Language (QL), Data Control Language (DCL)

Set Operations: Logical operators (NOT, IN, ALL, ANY, EXIST, NOT EXIST, LIKE, NOT LIKE, IS NULL, IS NOT NULL, AND, OR, NOT)

SQL Functions: Number Functions, Character Functions, Date Functions, Aggregate Functions

Integrity Constraints: Advantages of integrity constraints, Primary key, Unique key, Super key, Candidate key, Composite key, Foreign key, Domain constraints, Key constraints, Views in SQL, specifying general constraints and assertions